

Giacomo Cappelletto

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EDUCATION

Boston University

B.S. in Computer Engineering; GPA: 3.97/4.00

Expected May 2028

Boston, MA

RESEARCH & EXPERIENCE

Boston University - Prof. Brian Kulis

Undergraduate Machine Learning Researcher

Aug 2026 – Present

Boston, MA

- Researching robust contextual metric learning for human-motion retrieval by learning fixed-length embeddings from temporal 2D/3D pose sequences.
- Comparing contextual, contrastive, triplet, supervised-contrastive, and classification objectives with Recall@K and mAP under pose noise, missing joints, jitter, occlusion, and viewpoint shift.

Boston University College of Engineering

Undergraduate Machine Learning Researcher

Jan 2026 – May 2026

Boston, MA

- Built a Python vision pipeline for stabilization, single-athlete MMPose inference, MotionBERT 3D lifting, handle and machine tracking, stroke segmentation, and per-frame joint-angle extraction.
- Matched 20 controlled-run strokes into 1,203 aligned rows with 11.43 ms drive-duration MAE and trained sequence models over a shared 64-by-14 kinematic tensor for force-curve inference.

Banca Mediolanum

Software & AI Engineering Intern, Customer Knowledge

Jun 2026 – Aug 2026

Milan, Italy

- Built and deployed a LangGraph/MLflow agent with governed tools, retrieval, typed interfaces, evidence review, evaluation, and tracing on Databricks Model Serving.
- Reduced a representative SQL audit path by 79.2% while maintaining 1,200+ named tests across the agent and interface source snapshots.

Societa Cappelletto S.r.l.

Software Engineer

May 2025 – Oct 2025

Treviso, Italy

- Rebuilt a production inventory application in Tauri, React, and Rust, reducing size by 70%, memory use by 80%, startup time by 60%, and search latency by 43%.

SELECTED PROJECTS

Rowing Biomechanics Pipeline | GitHub

PyTorch, MMPose, MotionBERT

- Built a complete video-to-force research system with calibrated temporal alignment, quality-gated datasets, TCN/Transformer training, portable model bundles, video-only inference, and generated reports.

Deskinator - Autonomous Desk-Cleaning Robot | GitHub

Python, Raspberry Pi

- Built APDS9960 edge sensing, stepper odometry, RANSAC rectangle fitting, boustophedon coverage planning, telemetry, and simulation-based evaluation.

LEADERSHIP & ATHLETICS

Boston University Men's Rowing | NCAA Division I Student-Athlete | Sep 2024 – Present

- Freshman Student-Athlete of the Year (2025); Most Improved Oarsman (2026).

SKILLS

Languages: Python, C/C++, Rust, SQL, TypeScript/JavaScript

ML/CV: PyTorch, TensorFlow/Keras, MMPose, MotionBERT, metric learning, contrastive learning, pose estimation, 3D motion, sequence modeling, RANSAC

Platforms: MLflow, Databricks, LangGraph, PostgreSQL, Docker, Google Cloud Run, Git